

ACC 2026 Abstract Tiering Experiment

Cardiovascular Therapeutic Area | Synthetic Abstracts | Merck Data Tiering Criteria (2025)

This document contains Merck's official data tiering criteria, relevant CV portfolio context, and 12 congress abstracts. The goal is to evaluate how well AI can replicate the manual tiering process that EDSAs currently perform by hand.

Merck and MSD Data Tiering Criteria (Updated 2025)

Tier 1

- Phase 3 first primary endpoint, additional primary endpoint, or interim analysis (IA)
- Phase 2 that are practice-changing and/or have a path to accelerated approval
- New pipeline/assets data with standout efficacy/safety signals in Phase 1-2 studies
- Impactful new pipeline and assets data

Tier 2

- Data in which teams need to be prepared to respond
- Additional data analysis for Phase 3 studies that may represent relevant further evidence of efficacy/safety in ITT and/or cohort group
- Impactful non-interventional data

Tier 3 and TIP

- All other Merck data where enterprise-level response is not necessary, including trials in progress (TIP)

Notes:

- Tier 1 data to be highlighted in a press release will be confirmed by the steering committee.
- Alliance and external collaboration data (determination of internal or competitor nature, tiering and company response) will be considered case by case.
- "Impactful" may be positive or negative data.

Tier 1 vs. Tier 2 Activities (for context)

Tier 1 only: Study messages, consideration for press release, considerations for inclusion in WWCB, Verbal Response Documents (VRDs), Core Response Document (CRD).

Both Tier 1 and Tier 2: Study statements, earned media, thought leadership, sponsored content, digital/social media.

Tier 2 only: Consideration for inclusion in curtain raiser when available. Under exceptional circumstances: consideration for CRD and VRD.

Merck CV Portfolio Context

Marketed Products

Verquvo (vericiguat) - sGC stimulator for heart failure with reduced ejection fraction (HFrEF). Approved based on VICTORIA trial. Competes with SGLT2 inhibitors (empagliflozin/Jardiance, dapagliflozin/Farxiga) and ARNIs (sacubitril/valsartan/Entresto).

Pipeline Assets

Enlicitide (MK-0616) - Oral PCSK9 inhibitor for hyperlipidemia/ASCVD. Phase 2b completed, Phase 3 CVOT underway. Competes with injectable PCSK9 inhibitors (evolocumab/Repatha, alirocumab/Praluent), inclisiran/Leqvio (siRNA, twice-yearly), and bempedoic acid/Nexletol.

Congress Abstracts to Tier

Review each abstract below and assign a tier based on the criteria in Section 1. Consider the company (Merck vs. competitor), the phase, the type of data (primary endpoint vs. subgroup vs. design-only), and the competitive relevance to Merck's CV portfolio.

ACC-001: VICTOR-HF: Vericiguat Reduces Cardiovascular Death and Heart Failure Hospitalization in Patients with HFpEF — Primary Results of a Phase 3 Randomized Trial

Session Type: Late-Breaking Clinical Trial | **Authors:** Butler J, Anker SD, Packer M, et al. | **Company:** Merck | **Product:** Verquvo (vericiguat) | **Mechanism of Action:** sGC stimulator | **Disease Area:** Heart Failure with Preserved Ejection Fraction (HFpEF) | **Phase:** Phase 3

Background: Vericiguat improved outcomes in HFrEF (VICTORIA). Whether benefit extends to HFpEF is unknown. Methods: 5,200 patients with HFpEF (LVEF $\geq$ 45%) randomized to vericiguat 10mg or placebo. Primary endpoint: composite of CV death or first HF hospitalization. Results: Over median 28 months, primary endpoint occurred in 18.2% (vericiguat) vs 21.6% (placebo); HR 0.82, 95% CI 0.73-0.92, $p=0.0008$. CV death: HR 0.85 (0.72-1.00). First HF hospitalization: HR 0.79 (0.68-0.91). Safety: symptomatic hypotension 9.1% vs 5.8%. Conclusion: Vericiguat significantly reduced CV death and HF hospitalization in HFpEF, establishing a new treatment option for this underserved population.

ACC-002: MK-0616 (Enlicitide) Phase 2b Dose-Ranging: Oral PCSK9 Inhibition Achieves 60% LDL-C Reduction with Favorable Tolerability

Session Type: Late-Breaking Clinical Trial | **Authors:** Ballantyne CM, Bays HE, Catapano AL, et al. | **Company:** Merck | **Product:** Enlicitide (MK-0616) | **Mechanism of Action:** PCSK9 inhibitor (oral) | **Disease Area:** Hyperlipidemia / Atherosclerotic Cardiovascular Disease | **Phase:** Phase 2b

Background: Injectable PCSK9 inhibitors effectively lower LDL-C but oral alternatives could improve adherence. MK-0616 is a novel oral PCSK9 inhibitor. Methods: 1,200 patients with hyperlipidemia on maximally tolerated statins randomized to MK-0616 10mg, 30mg, 100mg, or placebo daily for 24 weeks. Primary endpoint: percent change in LDL-C at week 12. Results: LDL-C reduction from baseline: 38% (10mg), 52% (30mg), 60% (100mg) vs 2% (placebo); all $p<0.0001$. PCSK9 free levels reduced $>90\%$ at 30mg and 100mg. AEs similar across groups; GI events 8% vs 5% placebo. No hepatotoxicity signals. Conclusion: Oral MK-0616 produced robust, dose-dependent LDL-C lowering comparable to injectable PCSK9 inhibitors with a favorable safety profile, supporting Phase 3 development.

ACC-003: Empagliflozin Reduces Major Adverse Cardiovascular Events in Patients with HFmrEF: Primary Results from EMPEROR-Intermediate

Session Type: Late-Breaking Clinical Trial | **Authors:** Packer M, Anker SD, Butler J, et al. | **Company:** Boehringer Ingelheim / Eli Lilly | **Product:** Jardiance (empagliflozin) | **Mechanism of Action:** SGLT2 inhibitor | **Disease Area:** Heart Failure with Mildly Reduced Ejection Fraction (HFmrEF) | **Phase:** Phase 3

Background: SGLT2 inhibitors have demonstrated benefit in HFrEF and HFpEF. EMPEROR-Intermediate tested empagliflozin specifically in HFmrEF (LVEF 40-49%). Methods: 4,800 patients randomized to empagliflozin 10mg or placebo. Primary endpoint: composite of CV death or HF hospitalization. Results: Primary endpoint: 14.8% vs 17.9%; HR 0.80 (0.71-0.90), $p=0.0003$. Results consistent with prior EMPEROR trials. Conclusion: Empagliflozin reduces CV events across the full EF spectrum in heart failure.

ACC-004: Subgroup Analysis of VICTORIA: Vericiguat Efficacy by Baseline NT-proBNP Tertiles in Worsening Heart Failure

Session Type: Moderated Poster | **Authors:** Ezekowitz JA, Mebazaa A, Packer M, et al. | **Company:** Merck | **Product:** Verquvo (vericiguat) | **Mechanism of Action:** sGC stimulator | **Disease Area:** Heart Failure with Reduced Ejection Fraction (HFrEF) | **Phase:** Phase 3

Background: In VICTORIA, vericiguat reduced CV death and HF hospitalization in worsening HFrEF. Whether baseline natriuretic peptide levels modify treatment effect is clinically relevant. Methods: Post-hoc analysis of 5,050 VICTORIA patients stratified by baseline NT-proBNP tertiles (T1: <2000 , T2: 2000-5314, T3: >5314 pg/mL). Results: HR for primary endpoint: T1: 0.82 (0.67-1.00), T2: 0.88 (0.74-1.04), T3: 0.96 (0.82-1.12); p -interaction=0.24. Absolute risk reduction greatest in T1 (4.2%) vs T3 (1.1%). Conclusion: Vericiguat benefit was consistent across NT-proBNP subgroups without significant interaction, though absolute benefit was numerically greater in lower tertiles.

ACC-005: Real-World Persistence and Outcomes with Vericiguat in US Heart Failure Patients: A Retrospective Cohort Analysis

Session Type: Poster | **Authors:** Fonarow GC, Yancy CW, Hernandez AF, et al. | **Company:** Merck | **Product:** Verquvo (vericiguat) | **Mechanism of Action:** sGC stimulator | **Disease Area:** Heart Failure | **Phase:** Post-marketing

Background: Real-world evidence on vericiguat persistence and outcomes is limited. Methods: Retrospective analysis of 12,400 HFrEF patients prescribed vericiguat (2022-2025) from Optum EHR data. Primary outcome: 12-month persistence. Secondary: composite of all-cause death or HF hospitalization. Results: 12-month persistence: 62%. Median time to discontinuation: 14 months. Among persistent users, all-cause death or HF hospitalization was 24.3% vs 31.8% in non-persistent (adjusted HR 0.72, 0.64-0.81). Common discontinuation reasons: hypotension (18%), cost (15%), clinical improvement (12%). Conclusion: Real-world vericiguat persistence was moderate with significant outcome benefit among adherent patients.

ACC-006: Bempedoic Acid Reduces MACE in Statin-Intolerant Patients with CKD Stage 3: Prespecified CLEAR Subgroup Analysis

Session Type: Moderated Poster | **Authors:** Nissen SE, Lincoff AM, Brennan D, et al. | **Company:** Esperion Therapeutics | **Product:** Nexletol (bempedoic acid) | **Mechanism of Action:** ACL inhibitor | **Disease Area:** Atherosclerotic Cardiovascular Disease / Chronic Kidney Disease | **Phase:** Phase 3

Background: CLEAR demonstrated bempedoic acid reduces MACE in statin-intolerant patients. CKD patients are at elevated CV risk and often statin-intolerant. Methods: Prespecified analysis of 3,200 CLEAR participants with eGFR 30-59 mL/min/1.73m². Primary endpoint: 4-point MACE. Results: Bempedoic acid reduced MACE by 18% (HR 0.82, 0.70-0.96, p=0.01) in the CKD subgroup, consistent with the overall trial. LDL-C reduction: 22% vs placebo. eGFR stable over 40 months. Conclusion: Bempedoic acid is effective and kidney-safe in statin-intolerant CKD patients.

ACC-007: Design and Rationale of MK-0616-017: A Phase 3 Cardiovascular Outcomes Trial of Oral Enlicitide in High-Risk ASCVD Patients

Session Type: Poster | **Authors:** Raal FJ, Kallend D, Ray KK, et al. | **Company:** Merck | **Product:** Enlicitide (MK-0616) | **Mechanism of Action:** PCSK9 inhibitor (oral) | **Disease Area:** Atherosclerotic Cardiovascular Disease | **Phase:** Phase 3

Background: MK-0616 (enlicitide) is a novel oral PCSK9 inhibitor that demonstrated robust LDL-C lowering in Phase 2. Methods: MK-0616-017 is a randomized, double-blind, placebo-controlled, event-driven Phase 3 CVOT. 13,000 patients with established ASCVD and LDL-C $\geq$ 70 mg/dL on maximally tolerated statins will be randomized 1:1 to enlicitide 100mg or placebo daily. Primary endpoint: time to first occurrence of 3-point MACE (CV death, nonfatal MI, nonfatal stroke). Expected event-driven completion: ~5 years. Secondary endpoints include individual MACE components, all-cause mortality, and coronary revascularization. Results: Enrollment began Q1 2026; 4,200 patients enrolled as of data cutoff. Conclusion: This large-scale CVOT will determine whether oral PCSK9 inhibition with enlicitide reduces cardiovascular events.

ACC-008: Patient-Reported Outcomes with Sacubitril/Valsartan vs Valsartan in Elderly HFpEF Patients: PARAGON-HF Extended Follow-Up

Session Type: Poster | **Authors:** Solomon SD, McMurray JJV, Claggett B, et al. | **Company:** Novartis | **Product:** Entresto (sacubitril/valsartan) | **Mechanism of Action:** ARNI (neprilysin inhibitor + ARB) | **Disease Area:** Heart Failure with Preserved Ejection Fraction | **Phase:** Phase 3

Background: PARAGON-HF narrowly missed its primary endpoint in HFpEF. Extended follow-up PRO data may inform clinical practice. Methods: KCCQ and EQ-5D collected at 6-month intervals through 48 months in 2,100 patients age $\geq$ 65. Results: Mean KCCQ-TSS improvement: 3.2 (sacubitril/valsartan) vs 1.8 (valsartan) at 48 months (p=0.04). EQ-5D VAS: similar between groups. Hospitalization-free days alive: 12.3 more with sacubitril/valsartan over 4 years (p=0.07). Conclusion: Extended follow-up suggests modest quality-of-life benefits with sacubitril/valsartan in elderly HFpEF, primarily driven by physical limitation improvements.

ACC-009: Prevalence and Predictors of Hyperkalemia in Heart Failure Patients Receiving RAAS Inhibitors and MRAs: Analysis from the PINNACLE Registry

Session Type: Poster | **Authors:** Desai AS, Liu J, Vaduganathan M, et al. | **Company:** Multiple / Registry | **Product:** N/A | **Mechanism of Action:** N/A (registry analysis) | **Disease Area:** Heart Failure | **Phase:** N/A

Background: Hyperkalemia limits optimal RAAS/MRA use in HF. Methods: Cross-sectional analysis of 340,000 HF patients in the PINNACLE registry (2019-2025). Results: Hyperkalemia ($K \geq 5.5$) prevalence: 8.4% overall, 14.2% with dual RAAS+MRA. Independent predictors: CKD (OR 2.8), diabetes (OR 1.6), dual RAAS+MRA (OR 2.1), age >75 (OR 1.4). Only 42% of patients with $K \geq 5.5$ had potassium management documented. Conclusion: Hyperkalemia remains common in HF and is associated with suboptimal neurohormonal blockade, highlighting unmet need for potassium management strategies.

ACC-010: Inclisiran Every 6 Months Maintains Durable LDL-C Lowering Through 4 Years: ORION-3 Open-Label Extension

Session Type: Oral Presentation | **Authors:** Ray KK, Wright RS, Kallend D, et al. | **Company:** Novartis | **Product:** Leqvio (inclisiran) | **Mechanism of Action:** siRNA targeting PCSK9 | **Disease Area:** Hyperlipidemia | **Phase:** Phase 3

Background: Inclisiran, a twice-yearly siRNA targeting hepatic PCSK9 synthesis, showed sustained LDL-C lowering through 3 years. Methods: ORION-3 open-label extension: 382 patients received inclisiran 300mg SC every 6 months for 4 years. Results: Mean LDL-C reduction from baseline maintained at 48-52% through 4 years. PCSK9 levels remained suppressed >80%. Adherence 97% (administered in office). No new safety signals; injection-site reactions 5%, all mild/moderate. Conclusion: Inclisiran provides durable, consistent LDL-C lowering over 4 years with excellent adherence through twice-yearly dosing.

ACC-011: Machine Learning-Based Prediction of 30-Day Heart Failure Readmission Using Wearable Device Data: A Prospective Validation Study

Session Type: Poster | **Authors:** Sharma A, Topol EJ, Steinhubl SR, et al. | **Company:** Academic / Device | **Product:** N/A | **Mechanism of Action:** N/A (digital health) | **Disease Area:** Heart Failure | **Phase:** N/A

Background: Wearable devices may detect HF decompensation before hospitalization. Methods: Prospective validation of an ML model using continuous HR, HRV, activity, and SpO2 from 1,800 HF patients wearing a smartwatch post-discharge. 30-day readmission prediction. Results: AUC 0.78 (95% CI 0.74-0.82). Sensitivity 71%, specificity 72%. Median alert lead time: 4.2 days before readmission. PPV 34%, NPV 93%. Conclusion: Wearable-based ML models show promise for early HF readmission detection but PPV needs improvement for clinical deployment.

ACC-012: Vericiguat Improves Exercise Capacity in Heart Failure: Pooled Analysis of Cardiopulmonary Exercise Testing from VICTORIA and VITALITY-HFpEF

Session Type: Oral Presentation | **Authors:** Armstrong PW, Pieske B, Anstrom KJ, et al. | **Company:** Merck | **Product:** Verquuvo (vericiguat) | **Mechanism of Action:** sGC stimulator | **Disease Area:** Heart Failure | **Phase:** Phase 3 (pooled)

Background: Exercise intolerance is a cardinal HF symptom. Vericiguat's effect on objective exercise measures across the EF spectrum has not been pooled. Methods: Patient-level pooled analysis from VICTORIA (HFrEF, n=780 CPET substudy) and VITALITY-HFpEF (n=520). Outcomes: peak VO2, 6-minute walk distance (6MWD), ventilatory efficiency (VE/VCO2 slope). Results: Vericiguat improved peak VO2 by 0.9 mL/kg/min (p=0.003) and 6MWD by 22 meters (p=0.001) vs placebo across both trials. VE/VCO2 slope improved in HFrEF (-1.4, p=0.01) but not HFpEF (-0.6, p=0.18). Improvement in peak VO2 correlated with NT-proBNP reduction (r=-0.32, p<0.001). Conclusion: Vericiguat consistently improves exercise capacity across the HF spectrum.

Tiering Results Table

For each abstract, assign a tier (Tier 1, Tier 2, or Tier 3), a confidence level (High / Medium / Low), and a brief rationale explaining which criterion drove the assignment.

ID	Title (short)	Company	Assigned Tier	Confidence	Rationale
ACC-001	VICTOR-HF: Vericiguat Reduces Cardiovascular Death and ...	Merck			
ACC-002	MK-0616 (Enlicitide) Phase 2b Dose-Ranging: Oral PCSK9 ...	Merck			
ACC-003	Empagliflozin Reduces Major Adverse Cardiovascular Even...	Boehringer Ingelheim / Eli Lilly			
ACC-004	Subgroup Analysis of VICTORIA: Vericiguat Efficacy by B...	Merck			
ACC-005	Real-World Persistence and Outcomes with Vericiguat in ...	Merck			
ACC-006	Bempedoic Acid Reduces MACE in Statin-Intolerant Patien...	Esperion Therapeutics			
ACC-007	Design and Rationale of MK-0616-017: A Phase 3 Cardiova...	Merck			
ACC-008	Patient-Reported Outcomes with Sacubitril/Valsartan vs ...	Novartis			
ACC-009	Prevalence and Predictors of Hyperkalemia in Heart Fail...	Multiple / Registry			
ACC-010	Inclisiran Every 6 Months Maintains Durable LDL-C Lower...	Novartis			
ACC-011	Machine Learning-Based Prediction of 30-Day Heart Failu...	Academic / Device			
ACC-012	Vericiguat Improves Exercise Capacity in Heart Failure:...	Merck			